

Samunnati: System Architecture Document

Anjan Mahapatra

Abstract

This document presents the comprehensive system architecture for **Samunnati**, India's leading agri-enterprise platform focused on FPO (Farmer Producer Organization) financing, market linkages, and data-driven advisory. The architecture is designed to support 2,200+ FPOs, 1.2 million+ farmers (growing to 5 million+ by FY30F), Rs.24,200 crore in AUM, and Rs.2,500+ crore in annual operating income by FY30F. The architecture follows a **microservices-based, cloud-native design** with offline-first capabilities for rural connectivity. Key components include: (1) multi-channel FPO/farmer onboarding system, (2) Unnati SaaS platform for FPO management, (3) credit scoring and underwriting engine, (4) NBFC lending and collections module, (5) output marketplace API integration layer, (6) parametric insurance distribution system, (7) data monetization platform, and (8) analytics and reporting layer. The architecture emphasizes scalability, fault tolerance, security, and compliance with RBI, DPDP Act, and NBFC regulations.

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1 Architecture Overview

Samunnati's system architecture is built on a **microservices-based, cloud-native design** deployed on AWS (Mumbai region) with edge nodes for offline operation in low-connectivity rural areas. The architecture supports the following key user personas:

- **Farmers (individual):** Receive loans (JLG model), track repayment schedule, access advisory, file insurance claims, view market prices
- **FPO board members/managers:** Onboard members, apply for loans, manage repayments, sell produce via marketplace, access Unnati SaaS platform
- **Mitra/Field officers:** Onboard FPOs, conduct due diligence, collect repayments, provide training
- **Buyers (NinjaCart, WayCool, processors):** Access FPO produce listings, place orders, integrate via API
- **Corporate users (agri-input companies, insurers, government):** Access data APIs, purchase insights, monitor portfolios
- **Samunnati admins:** Underwrite loans, approve FPOs, monitor KPIs, manage NBFC compliance

1.1 High-Level Architecture Diagram

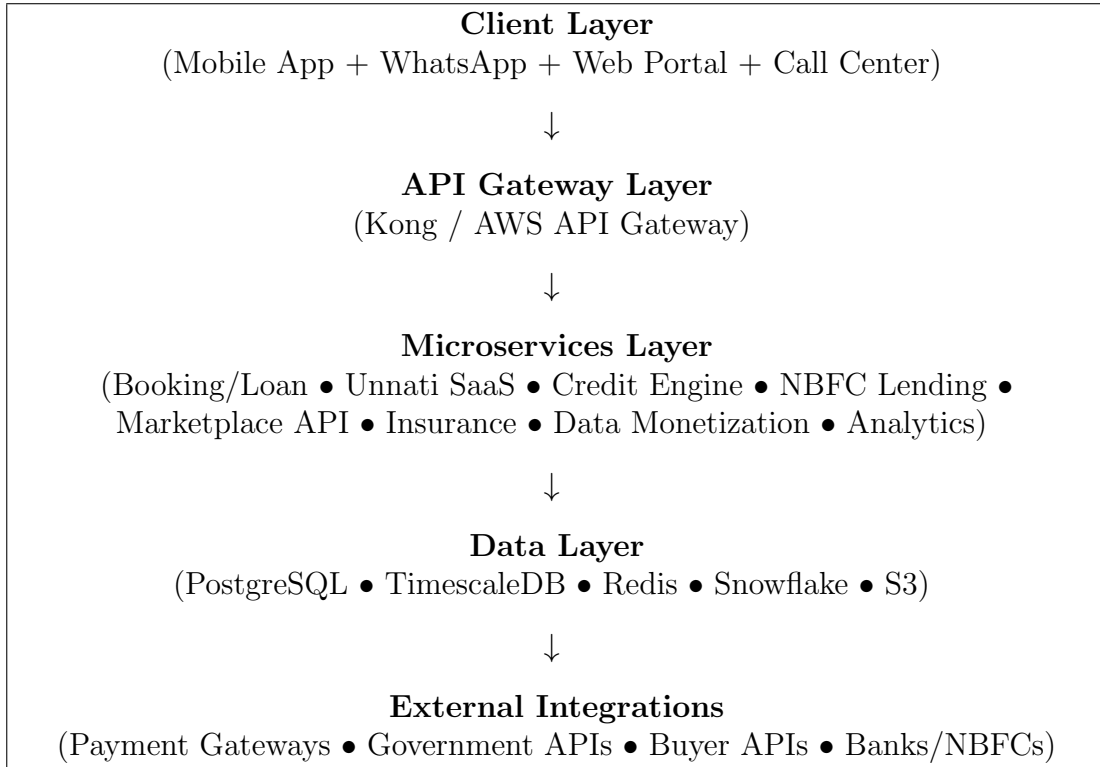


Figure 1: High-Level System Architecture

2 Client Layer

The client layer provides multi-channel access to Samunnati's services, catering to FPOs and farmers with varying levels of digital literacy and connectivity.

2.1 Mobile Application (Android & iOS)

- **Technology:** React Native (cross-platform), offline-first with SQLite local storage
- **Key features:**
 - Farmer-facing: Loan application status, repayment schedule, advisory (crop planning, pest alerts), market prices, insurance claim filing
 - FPO manager-facing: Member management, loan application, repayment tracking, produce listing on marketplace, Unnati dashboard
 - Field officer-facing: FPO onboarding checklist, due diligence forms, repayment collection, training materials
 - Offline mode: Forms stored locally; sync when connectivity returns
 - Multilingual support: Hindi, Tamil, Telugu, Kannada, Marathi, Gujarati, Bengali, Odia, English

2.2 WhatsApp Integration

- **Technology:** WhatsApp Business API (Meta)
- **Use cases:**
 - Loan status: Farmer sends "STATUS [loan ID]" → automated response with outstanding amount, next due date
 - Repayment reminder: Automated notifications 7 days, 3 days, 1 day before due date
 - Advisory: Daily crop-specific tips, weather alerts, pest warnings
 - Support: Raise tickets; relationship manager responds within 4 hours
- **Language support:** NLP for Hinglish and 8 regional languages

2.3 Web Portal (Unnati SaaS for FPOs & Corporate Users)

- **Technology:** React.js with TypeScript
- **FPO portal features:**
 - Member register management (Aadhaar-linked, biometric-enabled)
 - Loan application and tracking
 - Repayment scheduling and payment gateway integration
 - Produce listing on output marketplace (api-connected to NinjaCart, WayCool)

- GST/TDS compliance and automated filing
- Financial reporting (P&L, balance sheet for FPO)
- **Corporate portal features:**
 - API key management for data access
 - Data subscription management (tiered plans)
 - Portfolio analytics (for impact investors, government)

2.4 Call Center Integration

- **Technology:** Ozonetel / Exotel (cloud telephony)
- **Features:**
 - IVR (Interactive Voice Response) in 8 regional languages
 - Automated call-back for missed calls (farmer gives missed call → system calls back)
 - Agent dashboard with customer 360° view (loan history, repayment status, FPO membership)
 - Screen pop with AI-suggested responses based on farmer's past interactions

3 API Gateway Layer

The API Gateway serves as the single entry point for all client requests, providing security, routing, and rate limiting.

Table 1: API Gateway Configuration

Component	Specification
Technology	Kong / AWS API Gateway
Authentication	JWT + OTP (farmers); API keys (corporate)
Rate Limiting	100/min (farmer); 1,000/min (FPO); 10,000/min (corporate)
Request Routing	Endpoint prefix: /loans, /members, /marketplace, /data, /insurance
SSL/TLS	TLS 1.3 + auto cert renewal
DDoS Protection	AWS Shield + WAF
Logging	Centralized to ELK stack

4 Microservices Layer

The microservices layer comprises loosely coupled services that can be scaled independently.

4.1 Loan Origination & Underwriting Service

This service handles the complete loan lifecycle from application to disbursement.

- **Technology:** Java Spring Boot / Drools (rules engine)
- **Database:** PostgreSQL (read-write) + Redis (caching)
- **Key functions:**
 - FPO onboarding: Digital application with e-sign; Aadhaar-based KYC for members; land record upload
 - Credit underwriting: ML model using 50+ features (governance score, member profile, satellite NDVI, mandi prices)
 - Loan structuring: Tenor (6-24 months), interest rate (11-14%), repayment schedule (bullet or installment)
 - Disbursement: Direct to input suppliers (seeds, fertilizers) or FPO bank account with joint sign-off
 - Repayment tracking: Automated deduction from produce sale proceeds (integration with output marketplace)

4.1.1 Credit Scoring Model Features

Table 2: Credit Scoring Model Input Features

Category	Features
FPO Governance	Board composition, meeting frequency, audit trail, bank account discipline
Member Profile	Avg. landholding, cropping diversity, off-farm income, repayment history
Satellite Data	NDVI (crop health), soil moisture, historical yield
Market Data	Real-time mandi prices, buyer creditworthiness, export demand
External Data	CIBIL score (if available), monsoon forecast, pest surveillance

4.2 Unnati SaaS (FPO Management) Service

This service powers the Unnati platform used by FPOs for member management, accounting, and compliance.

- **Technology:** Python Django + Celery (async tasks)
- **Key functions:**
 - Member register: Aadhaar-linked, biometric-enabled, with crop and land records
 - Loan management: Application status, repayment schedule, interest calculation

- Accounting: Automated GST/TDS filing, integration with Tally/Zoho
- Compliance: PM-KISAN, PMFBY, and other government scheme reporting
- Buyer management: Off-take agreement templates, invoice generation, payment tracking

4.3 Marketplace API Integration Service

This service connects Samunnati-financed FPOs to output marketplaces (NinjaCart, WayCool, Jumbotail, AgriBazaar).

- **Technology:** Node.js + GraphQL
- **Key functions:**
 - Produce listing: Push FPO produce (quantity, quality grade, location, asking price) to multiple buyer platforms simultaneously
 - Order aggregation: Receive bids from multiple buyers; present best offer to FPO
 - Embedded finance: When buyer places order, offer instant working capital to FPO (repayment deducted from buyer payment)
 - Data sharing: Share FPO credit scores with buyers; share demand forecasts with FPOs

4.4 NBFC Lending & Collections Service

This service manages the NBFC loan book, collections, and regulatory reporting.

- **Technology:** Java Spring Boot + Quartz (scheduler)
- **Key functions:**
 - Loan servicing: Automated EMI deduction, late fee calculation, delinquency tracking
 - Collections: Automated SMS/WhatsApp reminders; field officer assignment for overdue accounts
 - Early warning system: ML model flags at-risk FPOs (payment behavior + crop health + market prices)
 - Regulatory reporting: Automated RBI reporting (NPA classification, capital adequacy, concentration limits)
 - Securitization: Pool loan data for securitization transactions; investor reporting

4.5 Parametric Insurance Service

This service enables farmers and FPOs to purchase crop insurance with automated payout.

- **Technology:** Node.js + integration APIs (ICICI Lombard, HDFC Ergo)
- **Key functions:**

- Quote generation: Real-time premium calculation based on crop, region, sum insured
- Policy issuance: Digital policy document with QR code for verification
- Automated payout: Trigger based on rainfall, temperature, or NDVI threshold breach; payout credited to loan account within 7 days
- Commission tracking: Auto-accrual of 5-10% commission; monthly payout

4.6 Data Monetization Service

This service sells anonymized, aggregated data to enterprise customers via API.

- **Technology:** Python FastAPI + Apache Spark (data aggregation)
- **Key functions:**
 - Data anonymization: Remove PII; aggregate at district/block level
 - API access: RESTful APIs with API keys; rate limiting by subscription tier
 - Report generation: Scheduled reports (daily, weekly, monthly) for government and agri-input companies
 - Data quality monitoring: Ensure 99.9% data accuracy; alert on anomalies
- **Datasets available:**
 - FPO credit performance by region, crop, governance score
 - Crop yield patterns by seed variety, input usage
 - Mandi price trends and forecasts
 - Equipment utilization (if telematics integrated)

4.7 Government Scheme Integration Service

This service integrates with government schemes (PM-KISAN, PMFBY, fertilizer subsidy) to process payments and claim fees.

- **Technology:** Node.js + API integrations (NPCI, DBT)
- **Key functions:**
 - Farmer enrollment: Link farmers to PM-KISAN, verify eligibility
 - Subsidy processing: Automatically credit subsidies to farmer loan account (reducing principal)
 - Scheme reporting: Generate reports for government on fund utilization
 - Fee income: Earn processing fee (1-2% of subsidy value)

4.8 Analytics & Reporting Service

This service provides business intelligence and operational dashboards.

- **Technology:** Apache Superset / Power BI + dbt (data transformation)
- **Key dashboards:**
 - Executive dashboard: AUM, GNPA, ROA, ROE, disbursements by region
 - Credit dashboard: Loan application volume, approval rate, average loan size, NPA trends
 - Collections dashboard: Repayment rate, delinquency aging, field officer performance
 - Marketplace dashboard: GMV facilitated, FPOs using API, embedded finance usage
 - Impact dashboard: Farmers served, women farmers percentage, income uplift

5 Data Layer

The data layer consists of multiple storage systems optimized for different use cases.

5.1 Operational Databases

Table 3: Operational Databases

Database	Technology	Purpose
Primary OLTP	PostgreSQL 15	FPOs, farmers, loans, repayments, members
Time-series	TimescaleDB	Telematics (GPS, engine hours - future), transaction logs
Geospatial	PostGIS	FPO hub locations, farmer clusters, distance-based search
Caching	Redis (cluster)	Session management, rate limiting, real-time dashboards
Search	Elasticsearch	FPO/farmer search, produce listings

5.2 Data Warehouse

- **Technology:** Snowflake / Amazon Redshift
- **Data sources:** All microservices stream data to Kafka → transformed via dbt → loaded to warehouse
- **Data volume:** 5 TB raw data per month (projected by FY30F)
- **Refresh frequency:** Real-time (streaming) for operational dashboards; daily for analytical reports

5.3 Data Lake

- **Technology:** Amazon S3 (with Glacier for archival)
- **Content:**
 - Raw telemetry data (satellite NDVI, weather) - retained 90 days in hot storage, 7 years in cold storage
 - Farmer KYC documents (Aadhaar, land records) - encrypted at rest
 - FPO board meeting minutes and audit reports
 - System logs (retained 30 days)

5.4 Data Governance & Security

- **Encryption:** AES-256 at rest (S3, RDS); TLS 1.3 in transit
- **Access control:** Row-level security in PostgreSQL; IAM roles for AWS services
- **Compliance:** RBI NBFC regulations, DPDP Act (India), ISO 27001 (planned FY27)
- **Backup:** Automated daily snapshots; cross-region replication to ap-south-1 (Mumbai) and ap-southeast-1 (Singapore)

6 External Integrations

Samunnati integrates with several external systems to enable seamless operations.

6.1 Payment Gateways

Table 4: Payment Gateway Integrations

Gateway	Use Case
Razorpay	UPI, credit/debit cards, net banking for FPO repayments
Cashfree	Payouts to FPOs (sale proceeds), commissions to field officers
Bank API (ICICI/HDFC/SBI)	NBFC loan disbursement and collection; NACH mandates

6.2 Government Systems

- **Aadhaar eKYC:** UIDAI API for farmer and FPO board member identity verification
- **DigiLocker:** Land record verification for loan underwriting
- **GST portal:** Invoice validation for input purchases and produce sales
- **PM-KISAN:** API integration to verify farmer eligibility and track subsidy payments
- **PMFBY (crop insurance):** API integration for policy lookup and claim status
- **Agmarknet:** Real-time mandi price feed

6.3 Third-Party Services

- **Google Maps API:** FPO hub location, farmer distance calculation
- **Satellite data APIs:** Sentinel-2, Landsat (NDVI, soil moisture) via AWS Ground Station
- **Weather APIs:** IMD (India Meteorological Department) for rainfall, temperature forecasts
- **Twilio:** SMS notifications (fallback when WhatsApp unavailable)
- **Sentry:** Error tracking and performance monitoring
- **Datadog:** Infrastructure monitoring and alerting

7 Scalability & Performance

7.1 Scaling Strategy

Table 5: Horizontal Scaling Targets by FY30F

Component	FY26F	FY30F
Active FPOs	800	2,200
Farmers covered (million)	1.6	5.0
Daily loan applications	500	2,000
Daily API requests (marketplace)	100,000	1,000,000
Data points processed/day (telematics, satellite)	10M	100M
Unnati SaaS active users	600	2,000

7.2 Auto-scaling Configuration

- **Orchestration:** EKS (Amazon Elastic Kubernetes Service) with Horizontal Pod Autoscaler (HPA)
- **Trigger conditions:** CPU >70%, memory >80%, custom metric (requests per second >1,000)
- **Node groups:**
 - Compute-optimized (c5.large): Loan origination, credit engine
 - Memory-optimized (r5.large): Redis caching, PostgreSQL
 - GPU-optimized (g4dn.xlarge): ML model inference (credit scoring, satellite data processing)

7.3 Performance SLAs

Table 6: Performance Service Level Agreements (SLAs)

Service	Target Response Time
Loan application submission (p95)	<1 sec
Credit decision (automated)	<10 sec
Produce listing on marketplace	<500 ms
Unnati dashboard load (p95)	<2 sec
API data access (aggregated)	<1 sec
API data access (raw export)	<5 sec

8 Deployment Architecture

8.1 AWS Infrastructure

- **Primary region:** ap-south-1 (Mumbai) - all production workloads
- **DR region:** ap-southeast-1 (Singapore) - cross-region replication for critical data (loan book, farmer profiles)
- **Edge locations:** CloudFront CDN with 10+ edge locations in India for static content (Unnati assets, app downloads)

8.2 Disaster Recovery

Table 7: Disaster Recovery Objectives

Metric	Target
Recovery Time Objective (RTO)	4 hours
Recovery Point Objective (RPO)	15 minutes (critical data: 5 minutes)
Backup frequency	Daily automated snapshots
DR testing frequency	Quarterly (simulated region failover)

9 Offline-First Design

Given that 30-40% of farmers and FPOs operate in areas with intermittent connectivity, Samunnati implements offline-first design principles.

9.1 Offline Capabilities by Component

Table 8: Offline Functionality

Component	Offline	Sync Strategy
FPO member onboarding	Yes	Local SQLite → batch sync
Loan application (draft)	Yes	Store locally → submit online
Repayment (field officer cash)	Yes	Sync device → upload within 24h
Produce listing (marketplace)	No	N/A
Crop advisory (cached)	Yes	Pre-download daily; refresh online
Unnati dashboard	Partial (24h cache)	Sync on app open
Insurance claim (photos)	Yes	Upload on connectivity

9.2 Sync Conflict Resolution

- **Last-write-wins:** For non-critical fields (e.g., profile preferences)
- **Server wins:** For financial transactions (loan applications, repayments)
- **Manual resolution:** For disputed loan collections or produce sale disputes
- **Idempotency keys:** Prevent duplicate loan applications when retrying failed API calls

10 Security Architecture

10.1 Authentication & Authorization

- **Farmers:** OTP-based authentication (mobile number + Aadhaar)
- **FPO managers:** JWT with role-based access (admin, treasurer, member)
- **Field officers:** JWT with geo-attendance validation
- **Corporate users:** API keys with IP whitelisting
- **Admins:** Multi-factor authentication (MFA) mandatory

10.2 Security Controls

Table 9: Security Controls Summary

Control	Implementation
Authentication	OTP (farmers); JWT + MFA (admins)
Authorization	RBAC, 8 roles (farmer, FPO admin, field officer, credit officer)
API Security	JWT validation at gateway; rate limiting by role
Data Encryption	AES-256 (rest); TLS 1.3 (transit)
Secrets Management	AWS Secrets Manager (API keys, DB credentials)
Vulnerability Scanning	Weekly automated (AWS Inspector, Snyk)
Penetration Testing	Annual third-party (CERT-In empaneled)
Audit Logging	All API calls logged (user ID, timestamp, IP)

10.3 RBI & DPDP Act Compliance

- **Data localization:** All farmer and FPO data stored only in India (ap-south-1 region)
- **Consent management:** Explicit consent for data collection (Aadhaar, biometrics); granular opt-out for marketing
- **Data retention:** PII deleted 7 years after last transaction (as per RBI NBFC guidelines)
- **Right to be forgotten:** API endpoint for account deletion (90-day processing time)
- **Data breach notification:** 72-hour notification to affected users, RBI, and CERT-In
- **NBFC reporting:** Automated RBI reporting (NPA classification, capital adequacy, concentration limits)

11 Monitoring & Observability

11.1 Monitoring Stack

Table 10: Monitoring Tools

Tool	Purpose
Datadog	Infrastructure metrics (CPU, memory, network, disk)
Prometheus + Grafana	App metrics (request rate, error rate, latency, loan approval rate)
ELK Stack	Centralized logging (application logs, audit logs)
AWS X-Ray	Distributed tracing for microservices
Sentry	Error tracking and crash reporting
PagerDuty	On-call alerting

11.2 Key Metrics to Monitor

Business KPIs:

- Loan disbursement volume (daily, weekly, monthly by region)
- GNPA and NNPA trends
- Repayment rate (on-time vs. delinquent)
- FPO onboarding rate and retention
- Marketplace GMV facilitated

Technical KPIs:

- API p95 latency by endpoint
- Error rate (HTTP 5xx) - target $\leq 0.1\%$
- Credit engine inference latency - target ≤ 500 ms
- Database connection pool utilization
- Kafka consumer lag (data pipeline health)

12 Cost Optimization

12.1 Current & Projected Cloud Spend

Table 11: Cloud Spend Projection (Rs. Cr)

Category	FY26F	FY28F	FY30F
Compute (EKS, EC2)	3.5	6.0	12.0
Storage (RDS, S3, Snowflake)	1.5	3.0	6.0
Data transfer	0.5	1.0	2.0
Third-party APIs (satellite, maps)	0.5	1.0	2.0
Total Cloud Spend	6.0	11.0	22.0

12.2 Optimization Strategies

1. **Reserved Instances:** Purchase 3-year reserved instances for stable workloads (30-40% savings on compute)
2. **Spot instances:** Use spot instances for non-production and batch processing (60-70% savings)
3. **S3 lifecycle policies:** Transition older data to Glacier after 90 days (80% storage cost reduction)
4. **CloudFront caching:** Cache Unnati dashboard assets and static content at edge (70% reduction in origin fetches)
5. **Data compression:** Compress telematics and satellite data before transmission (50% reduction in data transfer costs)
6. **Serverless for low-traffic services:** Use Lambda for infrequent operations (report generation, data exports) → pay-per-execution

13 Conclusion

The Samunnati system architecture is designed to support the company’s ambitious growth targets from Rs.138 crore (FY24A) to Rs.2,500+ crore (FY30F) while maintaining high availability (99.9% uptime target), sub-second response times for critical operations, and full compliance with RBI NBFC regulations and DPDP Act. The microservices-based, cloud-native design enables independent scaling of high-traffic components (loan origination, marketplace API, Unnati SaaS), while offline-first capabilities ensure usability in low-connectivity rural areas.

Key architectural decisions — including the choice of PostgreSQL with TimescaleDB for transaction processing, EKS (Kubernetes) for orchestration, Snowflake for analytics, and offline-first mobile apps — reflect the unique constraints of serving semi-literate, low-connectivity farmers and FPOs while managing high-volume loan transactions and real-time market linkages. The architecture is projected to handle 2,200+ FPOs, 5 million+ farmers, Rs.24,200 crore AUM, and 1 million daily API requests by FY30F.

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